higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

DIGITAL ELECTRONICS N4

14 August 2024

This marking guideline consists of 9 pages.

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SECTION A

QUESTION 1

1.1 C
1.2 A
1.3 D
1.4 D
1.5 B

(5 × 2) **[10]**

QUESTION 2

2.1 False
2.2 True
2.3 True
2.4 False
2.5 True
2.6 True
2.7 False
2.8 True

(8 × 1) **[8]**

QUESTION 3

3.1 F
3.2 E
3.3 C
3.4 B
3.5 G
3.6 A
3.7 D

(7 × 1) **[7]**

TOTAL SECTION A: 25

SECTION B

QUESTION 4

4.1

					BINARY								
POS					1	0	1,	0	1	1	1	1	
MIN				1	0	0	0,	1	0	0	1		
					BINARY								
POS		0*	0	0	1	0	1,	0	1	1	1	1	
MIN		1*	0	1	0	0	0,	1	0	0	1	0	
					1'S								
POS		0*	0	0	1	0	1,	0	1	1	1	1	√
MIN		1*	1	0	1	1	1,	0	1	1	0	1	√
	0	1*	1	1	1	0	0,	1	1	1	0	0	√
ADD												0	
		1*	1	1	1	0	0,	1	1	1	0	0	√
MIN			0	0	0	1	1,	0	0	0	1	1	√
MIN						1	1,	0	0	0	1	1	

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4.2

					BINARY									
MIN					1	1	0	1,	0	1	1			
MIN					1	0	1	0,	1	0	1	1		
					BINARY									
MIN			1*	0	1	1	0	1,	0	1	1	0		
MIN			1*	0	1	0	1	0,	1	0	1	1		
						2'S								
			1*	1	0	0	1	0,	1	0	1	0	√	
			1*	1	0	1	0	1,	0	1	0	1	√	
		1	1*	0	0	1	1	1,	1	1	1	1	√	
MIN				1	1	0	0	0,	0	0	0	0		
PLUS 1												1		
MIN				1	1	0	0	0,	0	0	0	1	√	√

4.3

							1	0	1	0,	0	1	1	1	
MULT									1	0	0,	1	0	1	
							1	0	1	0	0	1	1	1	√
					1	0	1	0	0	1	1	1			√
	1	0	1	0	0	1	1	1							√
EQU	1	1	0	0	0	0,	0	1	0	0	0	1	1	√	√

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4.4

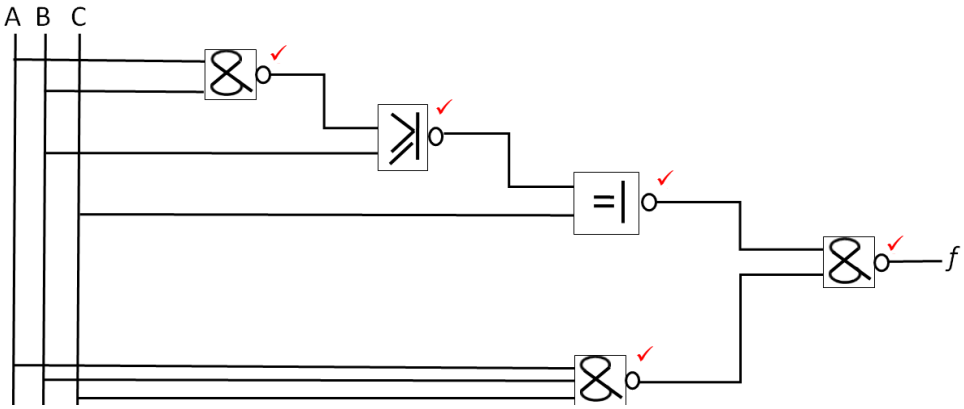
```
                                        1  0  0,  1  0  1   √ √
1  0  1  0,  0  1  1  1 | 1  1  0  0  0  0,  0  1  0  0  0  1  1
                        | 1  0  1  0  0  1  1  1
                        |          1  1  0  1  0  0  0  0      √
                        |          1  0  1  0  0  1  1  1
                        |             1  0  1  0  0  1  1  1   √
                        |             1  0  1  0  0  1  1  1
                                                   0          √
```

(4 × 5) **[20]**

QUESTION 5

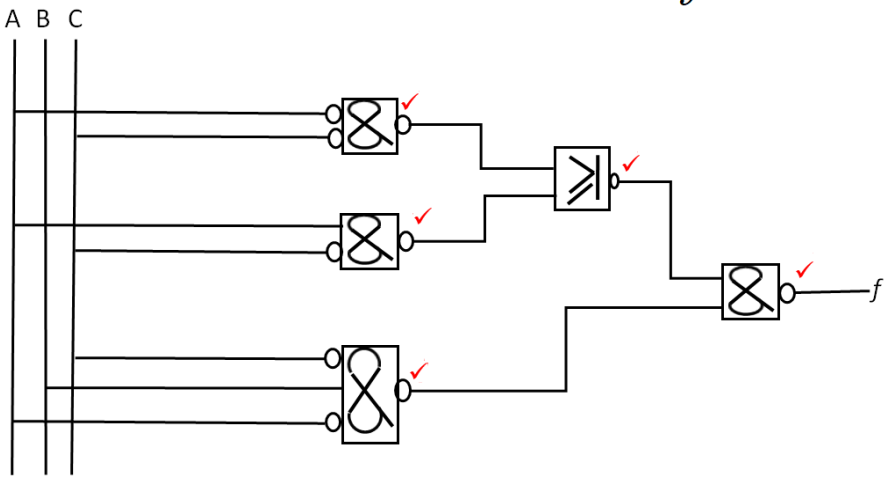
5.1

$$\overline{\overline{B+\overline{A\bullet B}\oplus C\bullet\overline{A\bullet B\bullet C}}}=f$$



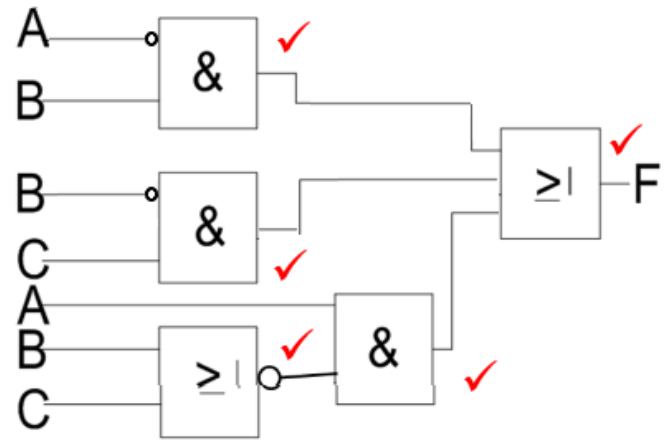
5.2

$$\overline{\overline{A\bullet\overline{C}}+\overline{A\bullet\overline{C}}\bullet\overline{\overline{A}\bullet B\bullet\overline{C}}}=f$$

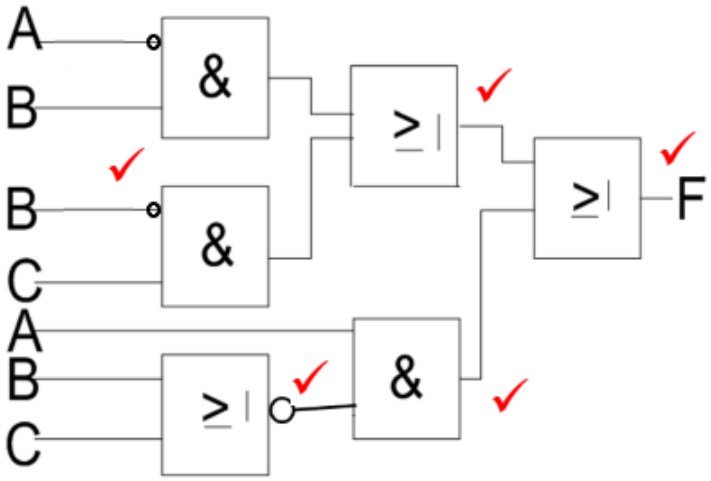


5.3

$$\overline{A}B+\overline{B}C+A(\overline{B+C})=f$$



OR



(3 × 5) **[15]**

QUESTION 6

$$A \otimes B$$

$$= \overline{A}B + A\overline{B} \quad \surd$$

$$= \overline{A}B + AB + A\overline{A} + B\overline{B}$$

$$= \overline{A}(A+B) + \overline{B}(A+B) \quad \surd$$

$$= (A+B) \bullet (\overline{A} + \overline{B}) \quad \surd$$

$$= (A+B) \bullet \overline{\overline{A} + \overline{B}}$$

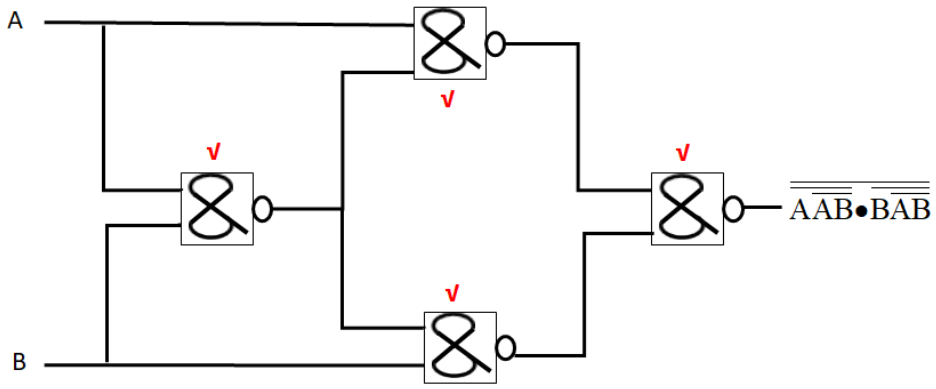
$$= (A+B) \bullet \overline{A \bullet B} \quad \surd$$

$$= (A+B) \bullet \overline{AB}$$

$$= A\overline{AB} + B\overline{AB} \quad \surd$$

$$= \overline{A\overline{AB} + B\overline{AB}}$$

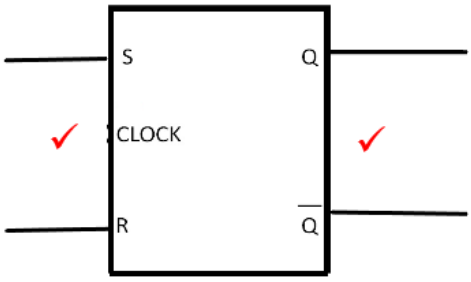
$$= \overline{A\overline{AB}} \bullet \overline{B\overline{AB}} \quad \surd$$



[10]

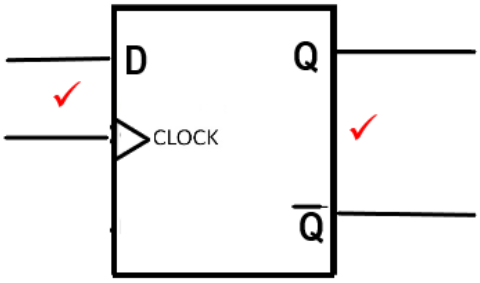
QUESTION 7

7.1 7.1.1



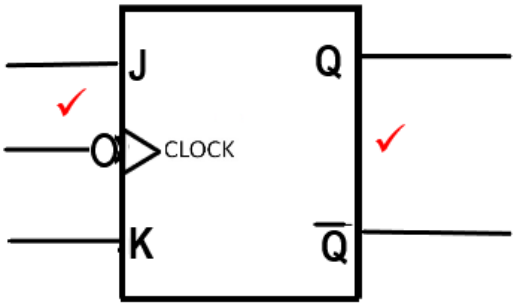
S	R	Q	$\overline{Q}$	
0	0	UNCHANGED		√
0	1	0	1	
1	0	1	0	√
1	1	UNDEFINED		

7.1.2



D	Q	$\overline{Q}$	
0	0	1	√
1	1	0	√

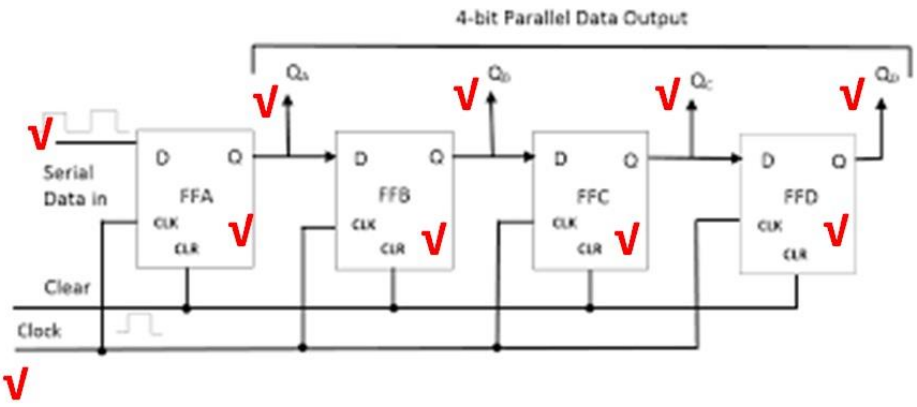
7.1.3



J	K	Q	$\overline{Q}$	
0	0	UNCHANGED		√
0	1	0	1	
1	0	1	0	√
1	1	TOGGLE		

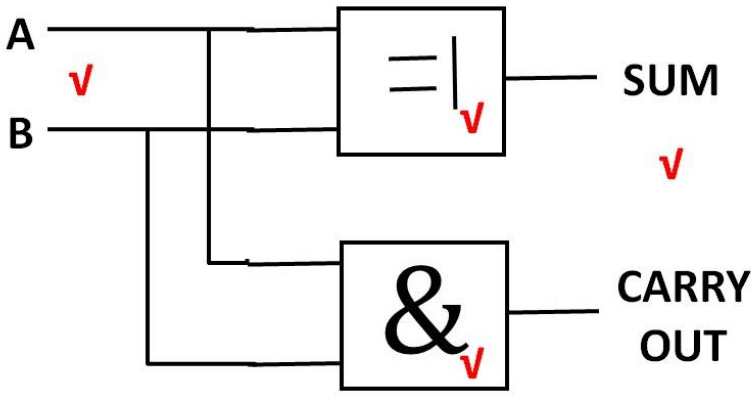
(3 × 4) (12)

7.2



(10)

7.3



A	B	SUM	C/OUT	
0	0	0	0	√
0	1	1	0	√
1	0	1	0	√
1	1	0	1	√

(8)
[30]

TOTAL SECTION B: 75
GRAND TOTAL: 100

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